Akriti Jain

Email | Website | Google Scholar | LinkedIn

Research Interests

My research focuses on evaluating and improving the reliability of language model outputs across objective and subjective tasks. I am particularly interested in understanding the interplay between social and epistemic alignment in real-world user interactions, with the goal of fostering calibrated trust.

Education

Cornell University — **Aug 2026 – Present**
Ph.D. in Computer Science | Advised by: Prof. Kaitlyn Zhou — *Ithaca, New York*

Indian Institute of Technology Roorkee — **Oct 2020 – May 2024**
B.Tech in Electrical Engineering (Gold Medalist) | Class Rank: 1/175 | GPA: 9.348/10.00 — *Roorkee, India*

Publications

- **[ACL 2026] Akriti Jain**, Anish Mulay, Divyansh Verma, Aishani Pandey, Pritika Ramu, Aparna Garimella. *Decisive: Guiding User Decisions with Optimal Preference Elicitation from Unstructured Documents.* [LINK]
- **[ACL Findings 2026]** Chau Minh Pham, Zichao Wang, Puneet Mathur, Alexa Siu, **Akriti Jain**, Aparna Garimella, Ananya B. Sai, Nedim Lipka, Mohit Iyyer, Varun Manjunatha. *AnalystBench: Benchmarking professional long-form report generation with web-mined multimodal tasks.* [LINK]
- **[ACL Findings 2026]** Jeonghyun Park, Ingeol Baek, Seunghyun Yoon, Haeun Jang, Aparna Garimella, **Akriti Jain**, Nedim Lipka, Hwanhee Lee. *MARCH: Evaluating the Intersection of Ambiguity Interpretation and Multi-hop Inference.* [LINK]
- **[EACL 2026] Akriti Jain**, Aparna Garimella. *Knowing What's Missing: Assessing Information Sufficiency in Question Answering.* [LINK]
- **[IJCNLP-AACL 2025] Akriti Jain**, Aparna Garimella. *Modeling Contextual Passage Utility for Multihop Question Answering.* [LINK]
- **[EMNLP 2025] Akriti Jain**, Pritika Ramu, Aparna Garimella, Apoorv Saxena. *Doc2Chart: Intent-Driven Zero-Shot Chart Generation from Documents.* [LINK]
- **[EMNLP Findings 2025] Akriti Jain**[*], Saransh Sharma[*], Koyel Mukherjee, Soumyabrata Pal. *FiRST: Finetuning Router-Selective Transformers for Input-Adaptive Latency Reduction.* [LINK]

Experience

Graduate Researcher, Cornell University — **Aug 2026 – Present**
Advisor: Prof. Kaitlyn Zhou — *Ithaca, New York*

- Investigating how user framing and social cues influence LLM judgments across open-ended evaluation tasks.
- Designed controlled experiments varying perceived social closeness and formality to quantify shifts in model evaluations.

Research Associate, Multimodal Content Experiences Lab, Adobe Research — **June 2024 – July 2026**
Group: Document Experiences | Mentors: Dr. Aparna Garimella, Dr. Koyel Mukherjee — *Bangalore, India*

- Conducted research on factual grounding, retrieval-quality assessment and intent-based generation over large document collections, contributing to 7 papers (5 as first author) and 7 patent applications as an inventor.
- Designed checklist-based and elimination-style evaluation methods for long-form generative artifacts such as deep research reports and presentations, and adopted them across engineering teams to guide system development.
- Developed a rule-based token compression method that reduced latency and cost by 15% in the summarization pipeline.
- Delivered a talk on AI alignment, titled "Are LLMs inherently people-pleasers?" at the annual research off-site.

Research Intern, Multimodal Content Experiences Lab, Adobe Research — **May 2023 – July 2023**
Topic: Multimodal LLMs for Design Generation and Editing | Mentor: Dr. Apoorv Saxena — *Bangalore, India*

- Engineered a compressed JSON format for design templates, reducing token count by 7× while maintaining visual quality.
- Fine-tuned a LLaMA-based MLLM for design completion using BERT-like masking strategies on Adobe Express templates, achieving a 3× IoU improvement in bounding-box prediction and enabling text-prompt-based element suggestions.
- Received a selective Pre-Placement Offer (PPO) for performance, awarded to only two members of the intern cohort.

Selected Projects

Input-Adaptive Layer Skipping for Efficient LLM Inference *Adobe Research*

- Developed a lightweight framework for dynamically skipping transformer layers, achieving a 10–20% latency reduction.
- Designed sequence-level routers during prefill, ensuring KV-cache compatibility and negligible recomputation overhead.
- Integrated LoRA fine-tuning to mitigate any quality loss while preserving 80–95% of the base model's performance.

Assessing Information Quality for Robust Question Answering *Adobe Research*

- Fine-tuned a lightweight model to score passage utility based on inter-passage dependencies for multi-hop QA, using reasoning traces as training signal to improve reranking and answer generation over relevance-based methods.
- Designed a framework to determine whether source context is sufficient to answer a question by identifying information gaps through self-consistency, achieving a 12% improvement on inferential QA and over 90% accuracy on faithfulness benchmarks.

Command Recognition from Dysarthric Speech Using Quantized DNN (under Prof. Manoj Tripathy**)** *IIT Roorkee*

- Developed a command-recognition system for dysarthric speech by building and training a deep neural network.
- Achieved up to 79.31% accuracy for low-intelligibility speakers through speaker-adaptive transfer learning.
- Applied post-training quantization to compress the model to 492 KB for TinyML deployment on a Raspberry Pi with an I2S MEMS microphone, enabling a portable assistive device for real-time offline inference on resource-constrained hardware.

Honours and Awards

Awarded the **Cornell Graduate Fellowship**. 2026

Received the **Department Gold Medal** for highest academic standing in Electrical Engineering (EE) at IIT Roorkee. 2024

Received the **Durga Award** for outstanding overall performance among graduating UG women students at IIT Roorkee. 2024

Awarded the **Air Cmdr Shyam Chand Mehra Scholarship** for the highest CGPA in EE at IIT Roorkee. 2021–24

Patents

- **Akriti Jain**, Pritika Ramu, Aparna Garimella. *Generating Charts from Documents Based on User Intent and Document Data*. (US Patent App. P13973-US).
- **Akriti Jain**, Saransh Sharma, Koyel Mukherjee, Soumyabrata Pal, Aayush Acharya, Saud Iqbal. *Skipping Layers in Large Language Models Utilizing Layer-Specific Routers with Low Rank Adapters*. (US Patent App. P13522-US).
- **Akriti Jain**, Aparna Garimella. *A Method to Assess Information Sufficiency in Question Answering*. (US Patent App. P14529-US).
- **Akriti Jain**, Aparna Garimella. *A Method to Assess the Utility of Retrieved Passages for Multi-hop QA*. (US Patent App. P14579-US).
- Anish Mulay, Divyansh Verma, Aishani Pandey, **Akriti Jain**, Pritika Ramu, Aparna Garimella. *A Method to Guide User Decisions with Optimal Preference Elicitation from Unstructured Documents*. (US Patent App. P14551-US).
- **Akriti Jain**, Amit Sinha, Koyel Mukherjee, Pannaga Shivaswamy. *RQ-Conv: Improved Related Questions Generation Using Conversation History*. (US Patent App. P15098-US).
- **Akriti Jain**, Saransh Sharma, Navin Bahadur, Jayant Vaibhav Srivastava, Parth Shailesh Patel, et al. *Deep-Proof: Evidence-Based Research Planning and Evaluation for Deep Research Agents*. (US Patent App. P15168-US).

Teaching and Mentorship

Mentor: Adobe Research. Supervised 6 undergraduate interns and collaborated with 2 PhD interns on research projects during the summer internship season; co-led university collaborations with the University of Maryland and Chung-Ang University.

Undergraduate Teaching Assistant: *Introduction to C++ (EEN-103)*, IIT Roorkee. Guided more than 50 freshers through C++ fundamentals and object-oriented programming (OOP) concepts, and conducted regular weekly doubt-clearing sessions.

Student Mentor: Student Mentorship Program, IIT Roorkee. Provided guidance to 5 first-year undergraduate students through academic coursework, internship and placement preparation, and overall adjustment to university campus life.

Positions of Responsibility

Member, GradSWE, Cornell University: Contributed to initiatives promoting community building, networking, and professional development among women graduate students in engineering.

Secretary, IIT Roorkee Chess Club: Headed a team of 30+ members, fostered a collaborative environment to promote campus chess culture, and coordinated with chess clubs at other IITs to facilitate inter-college tournaments.

PR and Fundraising Volunteer, Girl Up Uthaan: Organized awareness and donation drives for underprivileged children with 20+ contributors and planned live sessions to promote mental health awareness and menstrual hygiene.